

Subseasonal Forecast Skill over India: A Two-Season Benchmark of AI-Based and Operational Models

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Abstract

[**TODO:** Final abstract pass. Skeleton below states the three findings; tighten to ~200 words and add one quantitative hook per finding.]

Subseasonal-to-seasonal (S2S) prediction at two-to-six-week leads is of high value for Indian agriculture and water management, yet precipitation skill at these leads is limited and the performance of new machine-learning (ML) forecast systems over India is largely undocumented. We benchmark six S2S systems—three ML-based (Spire AI-S2S, FuXi-S2S, DLESyM) and three operational dynamical (ECMWF, UKMO, NCEP)—for weekly precipitation, 500-hPa geopotential height (Z500), and 2-m temperature (T2M) over India across two contrasting seasons: the synoptically driven winter (JFM 2026, 90 daily initializations) and the convectively dominated summer monsoon (JJAS 2019, 17 common initializations). Forecasts are verified against ERA5 (Z500, T2M), ERA5 daily totals and IMD gridded rainfall (precipitation), using cosine-weighted ACC, RMSE, bias, and a common probabilistic framework (CRPSS, spread-skill ratio). We report three findings. (i) In winter, the ML system Spire AI-S2S is the most skillful system for precipitation at every lead (week-1 ACC 0.78 versus 0.73 for ECMWF) and uniquely retains Z500 skill to week 6, whereas the other two ML systems (FuXi-S2S, DLESyM) lose all Z500 skill—falling to negative ACC—by week 3–4. (ii) In the monsoon, this ML advantage disappears: every system, ML and dynamical alike, loses precipitation skill by week 3 (all-India ACC ≤ 0.06 at week 3). (iii) The deterministic ranking does not carry over to probabilistic skill: ECMWF leads precipitation CRPSS despite Spire’s higher ACC. These results give an early, honest assessment of where ML S2S systems help over India and where the problem remains hard for all approaches.

Keywords: subseasonal-to-seasonal prediction; machine learning; precipitation; India; monsoon; ensemble verification

1 Introduction

[**TODO:** Expand. Topic sentence + societal motivation (rabi crops, water, WDs, monsoon). Reuse prose from the prior draft’s Introduction.] Subseasonal-to-seasonal (S2S) prediction bridges medium-range weather and seasonal climate forecasting and is increasingly demanded for Indian agriculture, water resources, and disaster preparedness (Vitart et al., 2017; White et al., 2017; Mariotti et al., 2020).

[**TODO:** Expand. ML weather models now rival NWP at medium range; extension to S2S is recent (FuXi-S2S); regional precip skill over India undocumented. Reuse Intro ¶3.] Machine-learning (ML) forecast systems now match or exceed operational numerical weather prediction (NWP) at medium range (Pathak et al., 2022; Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024), and their extension to the subseasonal range has begun (Chen et al., 2024; Spire Global, 2025).

[**TODO:** Expand into the paper’s central idea. Key argument: a single season cannot separate "ML is better" from "ML is better at easy regimes." We test two regimes deliberately — winter (synoptic, WD-driven, higher predictability) vs monsoon (convective, lower S/N) — to find WHERE ML helps.] A single-season benchmark cannot distinguish a genuinely better forecast system from one that merely performs well in an easy regime. We therefore evaluate the same six systems across two physically contrasting seasons over the same domain.

We address four questions:

1. How does weekly skill for precipitation, Z500, and T2M decay with lead time (weeks 1–6) for each system, in winter and in the monsoon?
2. Do ML systems outperform operational dynamical systems over India, and does any advantage persist across both seasons?
3. Do the ML systems behave consistently with one another, or do they diverge at long lead?
4. Does deterministic skill translate into probabilistic (calibrated) skill?

[**TODO:** Close Intro with a one-paragraph roadmap of the paper.]

2 Data and Study Domain

2.1 Study domain and regions

[**TODO:** Domain box 5–38N, 65–100E; land points only; cosine weights; four IMD homogeneous regions (Northwest, Central, South Peninsula, East/NE). Reuse Domain subsection + Fig. domain from old paper. Add region map figure.]

2.2 Verification references

[**TODO:** Describe the three truth sources and WHY each is used.]

- **Precipitation, winter:** ERA5 daily totals (00–00 UTC), [**TODO:** cite ARCO-ERA5 carver2023; note IMD-2026 not yet available.]
- **Precipitation, monsoon:** IMD 0.25° gauge-based gridded rainfall (Pai et al., 2014) — an observational reference independent of any reanalysis.
- **Z500 and T2M:** ERA5 (Hersbach et al., 2020); for JJAS the local WeatherBench2 ERA5 archive (Rasp et al., 2024).

2.3 Forecast systems

[**TODO:** One short paragraph; details in Table 1 and Sec. 3.]

Table 1: Forecast systems evaluated. “Native res.” is each model’s own horizontal grid; all are regridded to the common 1.5° verification grid before scoring. CF = control, PF = perturbed member. “Both” = evaluated in JFM 2026 and JJAS 2019.

System	Type	Native res.	Members	Horizon	Seasons
Spire AI-S2S	ML (data-driven)	0.5°	mean+s.d.	46 d	JFM 2026
FuXi-S2S	ML (transformer)	1.5°	1CF+10PF	42 d	Both
DLESyM	ML (diffusion)	0.25°	80 PF	48 d	Both
ECMWF	Dynamical, coupled	Tco319 (~ 36 km)	100 PF	46 d	Both
UKMO GloSea6	Dynamical, coupled	N216 (~ 60 km)	1CF+3PF	60 d	Both
NCEP CFSv2	Dynamical, coupled	T126 (~ 100 km)	15 PF	44 d	Both

3 Forecast Systems

[**TODO:** One subsection each. Keep to 4–6 sentences: lineage, resolution, ensemble, how each variable is delivered + unit handling. Reuse ECMWF/NCEP/ Spire/FuXi text from old paper; write UKMO + DLESyM fresh (see notes).]

3.1 Spire AI-S2S

[**TODO:** ML, 0.5° , mean+s.d., 46 d, ERA5-initialized, daily TP (no unit conv).]

3.2 FuXi-S2S

[**TODO:** ML transformer, 1.5° , 1CF+10PF, 42 d, TP delivered as $\text{mm h}^{-1} \Rightarrow \times 24$.]

3.3 DLESyM

[**TODO:** Diffusion-based ML, 0.25° , 80 members, 48 d, 6-hourly $\Rightarrow$ daily-mean; z500 geopotential $\div g$; NO precipitation channel (Z500/T2M only). Note 3 corrupted stores excluded.]

3.4 ECMWF extended-range ensemble

[**TODO:** IFS, Tco319, 100PF, 46 d; cumulative TP $\Rightarrow$ de-accumulate.]

3.5 UKMO GloSea6

[**TODO:** MetUM, N216, 1CF+3PF, 60 d; cumulative TP; z500 already gpm; T2M broken (single lead) $\Rightarrow$ excluded from rankings.]

3.6 NCEP CFSv2

[**TODO:** T126, 15PF, 44 d; cumulative TP.]

4 Verification Framework

4.1 Weekly lead windows and aggregation

[**TODO:** Day-1 = first 24 h after init; W1=days1–7 ... W6=days36–42; weekly means; common 1.5° grid; cosine-lat weights; land mask.]

4.2 Anomalies and climatology

[**TODO:** ERA5 1990–2019 day-of-year climatology for Z500/T2M and ERA5-truth TP; IMD 1991–2020 daily climatology for IMD-truth TP. Common climatology removed identically from forecast and truth.]

4.3 Deterministic and probabilistic metrics

[**TODO:** Define ACC (cosine-weighted), RMSE, bias, MAE; then CRPS, CRPSS vs climatology, spread–skill ratio; Brier for TP exceedance events. Equations can be lifted from old paper Sec. Metrics.]

4.4 Initialization sets and matched comparison

[**TODO:** Explain the two init sets explicitly — this is a methods strength:]

- **JFM 2026:** 90 daily 00 UTC initializations (1 Jan–31 Mar 2026). [**TODO:** note W5/W6 sample shrinks as ERA5 truth ends 31 Mar.]
- **JJAS 2019:** 17 common Thursday initializations (6 Jun–26 Sep 2019) on which all five non-Spire models provide usable forecasts; this matched set is the basis of all cross-model JJAS comparisons. A 35-initialization operational subset (ECMWF, UKMO, NCEP) is used as a robustness check (Sec. 5.7).

5 Results

5.1 Overview: skill across variables and seasons

Figure 1 summarizes the whole benchmark: anomaly correlation versus lead week for all six systems, three variables, and both seasons. [**TODO:** Walk the reader through the figure in 4–5 sentences, foreshadowing the three findings (winter ML advantage; monsoon collapse; ML divergence).]

5.2 Winter (JFM 2026): the ML advantage

[**TODO:** Lead with precipitation. SPIRE best at every lead (W1 0.78 vs 0.73 ECMWF/UKMO, 0.62 NCEP; lowest RMSE 1.09 mm/day, near-zero bias -0.04). Then Z500: SPIRE holds 0.69–0.77 W3–W6 while FuXi/DLESyM go NEGATIVE. Then T2M: SPIRE dominant. Cite the three ACC tables + RMSE table.]

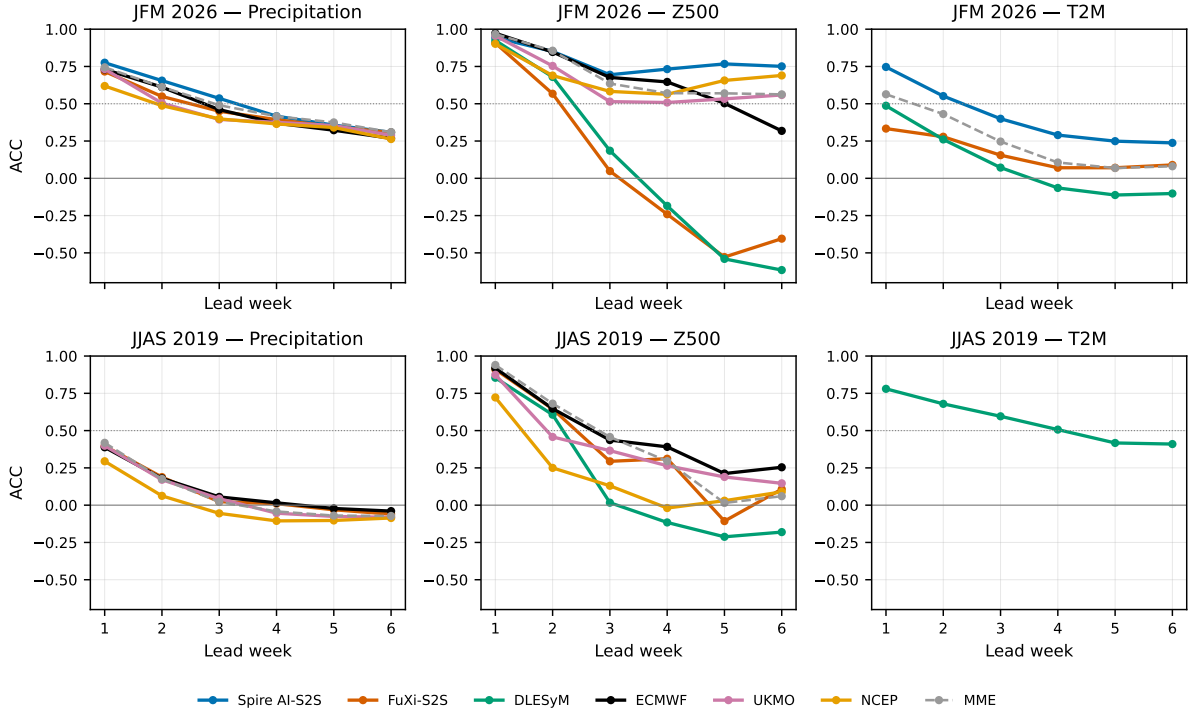


Figure 1: All-India anomaly correlation (ACC) versus lead week for six S2S systems. Top row: JFM 2026 (winter); bottom row: JJAS 2019 (monsoon). Columns: precipitation, Z500, 2-m temperature. Dotted line ACC= 0.5; solid line ACC= 0. DLESyM provides no precipitation; UKMO T2M and Spire (JJAS) are unavailable.

Table 2: JFM 2026 all-India precipitation anomaly correlation (ACC) by lead week, verified against ERA5 daily totals. Best individual-model score in each column is bold; MME is the multi-model mean.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.78	0.66	0.54	0.42	0.36	0.31
FuXi-S2S	0.72	0.55	0.45	0.39	0.35	0.31
ECMWF	0.73	0.61	0.46	0.37	0.32	0.27
UKMO	0.73	0.50	0.39	0.38	0.35	0.29
NCEP	0.62	0.49	0.40	0.36	0.34	0.26
MME	0.74	0.61	0.49	0.42	0.37	0.31

5.3 Monsoon (JJAS 2019): the great equalizer

[**TODO:** Key message: ALL systems collapse. Precip W1 0.4 (0.29 NCEP) -> <=0.06 by W3 -> negative after. No ML advantage. Z500 horizon also shorter than winter. DLESyM Z500 collapses fastest; ECMWF/UKMO most persistent. T2M only DLESyM available (0.78->0.41). Cite JJAS tables.]

5.4 Divergence among the ML systems

[**TODO:** Dedicated subsection making finding (iii) explicit: SPIRE vs FuXi/DLESyM. At W1 all three ML look comparable; by W3-4 SPIRE is flat while FuXi+DLESyM Z500 ACC < 0. Argue this is a long-lead stability / error-growth property, not a headline-score property — a caution against ranking ML systems on week-1 numbers alone. Reference Fig. 1 middle column.]

Table 3: JFM 2026 all-India Z500 ACC by lead week (ERA5 truth).

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.94	0.85	0.69	0.73	0.77	0.75
FuXi-S2S	0.91	0.57	0.05	-0.24	-0.53	-0.40
DLESyM	0.92	0.68	0.19	-0.18	-0.54	-0.62
ECMWF	0.97	0.85	0.68	0.65	0.50	0.32
UKMO	0.96	0.75	0.51	0.51	0.53	0.56
NCEP	0.90	0.69	0.58	0.56	0.66	0.69
MME	0.97	0.86	0.64	0.57	0.57	0.56

Table 4: JFM 2026 all-India 2-m temperature ACC by lead week (ERA5 truth).

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.75	0.55	0.40	0.29	0.25	0.24
FuXi-S2S	0.33	0.28	0.16	0.07	0.07	0.09
DLESyM	0.49	0.26	0.07	-0.06	-0.11	-0.10
MME	0.56	0.43	0.25	0.11	0.07	0.08

5.5 Probabilistic skill and calibration

[**TODO:** The "deterministic != probabilistic" finding. ECMWF leads TP CRPSS (0.59 W1) over SPIRE (0.61 W1 actually ties — CHECK and state precisely); for Z500 ECMWF leads early but SPIRE most persistent. Spread-skill: who is under/over-dispersive. Cite CRPSS tables + spread-skill figure.]

5.6 Regional structure of skill

[**TODO:** Use the scorecard: which IMD regions are most/least predictable in winter; does SPIRE's lead hold in all four regions. 3–4 sentences.]

5.7 Robustness: larger initialization set

[**TODO:** State that the JJAS conclusions are unchanged on the 35-init operational subset (ECMWF/UKMO/NCEP), confirming the monsoon collapse is not an artifact of the 17-init matched set. Optionally a small table or one sentence with numbers from the 35-init operational runs.]

6 Discussion

Where ML helps, and where it does not. [**TODO:** Synthesize findings (i)+(ii): ML's winter advantage is real but regime-dependent; the monsoon remains hard for all systems. Tie to predictability of synoptic (WD) vs convective regimes.]

Not all AI systems are equal. [**TODO:** Finding (iii): long-lead stability differs sharply across ML systems; caution against single-lead leaderboard rankings; hypothesize causes (training objective / error growth in transformer vs diffusion vs Spire).]

Table 5: JFM 2026 all-India precipitation RMSE (mm day^{-1}) by lead week.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	1.09	1.44	1.68	1.88	2.14	2.28
FuXi-S2S	1.33	1.83	1.95	2.08	2.34	2.47
ECMWF	1.27	1.59	1.92	2.14	2.40	2.57
UKMO	1.31	1.90	2.16	2.14	2.36	2.52
NCEP	1.57	1.93	2.15	2.29	2.49	2.72
MME	1.18	1.54	1.80	1.94	2.18	2.34

Table 6: JJAS 2019 all-India precipitation ACC by lead week, verified against IMD gridded rainfall (17 common initializations).

Model	W1	W2	W3	W4	W5	W6
FuXi-S2S	0.40	0.19	0.02	0.01	-0.03	-0.06
ECMWF	0.39	0.18	0.06	0.02	-0.02	-0.04
UKMO	0.40	0.17	0.04	-0.05	-0.08	-0.08
NCEP	0.29	0.06	-0.05	-0.11	-0.10	-0.09
MME	0.42	0.18	0.02	-0.04	-0.07	-0.07

Deterministic vs probabilistic skill. [**TODO:** Finding (iv): calibration is a distinct axis; ECMWF’s mature ensemble still leads probabilistically. Operational guidance should weigh both.]

Comparison with prior S2S literature. [**TODO:** Consistency with deandrade2019 / South-Asia S2S: precip skill to week 2; circulation horizon longer. Cite.]

7 Limitations

[**TODO:** Honest list: (a) two seasons / different years — not a multi-year hindcast; (b) JFM uses ERA5 precip truth (IMD-2026 unavailable) while JJAS uses IMD — truth differs between seasons; (c) Spire absent in 2019; (d) common ERA5 climatology, no per-model bias correction; (e) small ensembles for UKMO (4) limit its probabilistic scores; (f) UKMO T2M and DLESyM precip unavailable; (g) FuXi T2M has a known $\sim 4\text{K}$ cold offset (00 UTC snapshot vs daily-mean), reported but not corrected.]

8 Conclusions

[**TODO:** Restate the three+one findings crisply as numbered takeaways; end with the reproducible-framework sentence and outlook (more seasons, IMD-2026 truth, per-model climatology).]

Data and Code Availability

[**TODO:** ERA5/ARCO; IMD (Pai et al., 2014); ECMWF/UKMO/NCEP via S2S database (Vitart et al., 2017); FuXi-S2S weights (Zenodo/GitHub); Spire under research agreement; DLESyM via provider. Analysis code: <https://github.com/Artamta/s2s-analysis>.]

Table 7: JJAS 2019 all-India Z500 ACC by lead week (ERA5 truth, 17 common initializations).

Model	W1	W2	W3	W4	W5	W6
FuXi-S2S	0.91	0.64	0.29	0.31	-0.11	0.11
DLESyM	0.85	0.61	0.02	-0.12	-0.21	-0.18
ECMWF	0.92	0.65	0.44	0.39	0.21	0.25
UKMO	0.87	0.46	0.37	0.26	0.19	0.15
NCEP	0.72	0.25	0.13	-0.02	0.03	0.09
MME	0.94	0.68	0.46	0.30	0.01	0.06

Table 8: JFM 2026 all-India precipitation CRPSS versus climatology by lead week. Positive values indicate probabilistic skill above climatology.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.61	0.40	0.25	0.19	0.18	0.17
FuXi-S2S	0.45	0.27	0.31	0.32	0.29	0.26
ECMWF	0.59	0.51	0.40	0.35	0.30	0.25
UKMO	0.45	0.27	0.18	0.25	0.23	0.19
NCEP	0.42	0.33	0.29	0.26	0.24	0.15

Acknowledgements

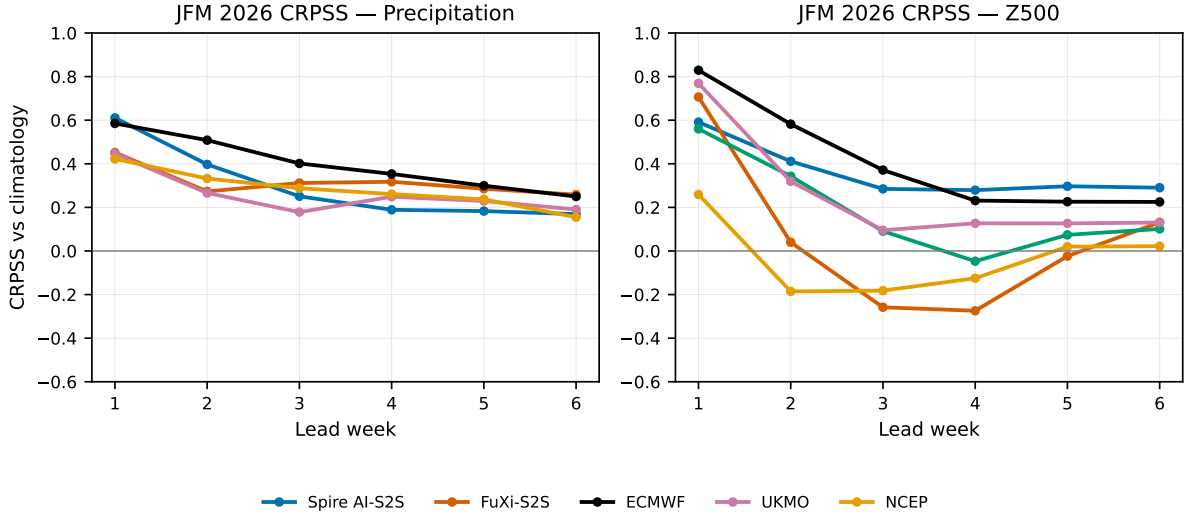
[**TODO:** Reuse from old paper; A.R.T. MSc thesis at IISER Pune; HPC; provider acknowledgements; LLM-assistance disclosure.]

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Table 9: JFM 2026 all-India Z500 CRPSS versus climatology by lead week.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.59	0.41	0.28	0.28	0.30	0.29
FuXi-S2S	0.71	0.04	-0.26	-0.27	-0.02	0.13
DLESyM	0.56	0.34	0.09	-0.05	0.07	0.10
ECMWF	0.83	0.58	0.37	0.23	0.23	0.23
UKMO	0.77	0.32	0.10	0.13	0.13	0.13
NCEP	0.26	-0.19	-0.18	-0.13	0.02	0.02

**Figure 2:** JFM 2026 all-India CRPSS versus lead week for precipitation (left) and Z500 (right). Positive values indicate probabilistic skill above climatology.

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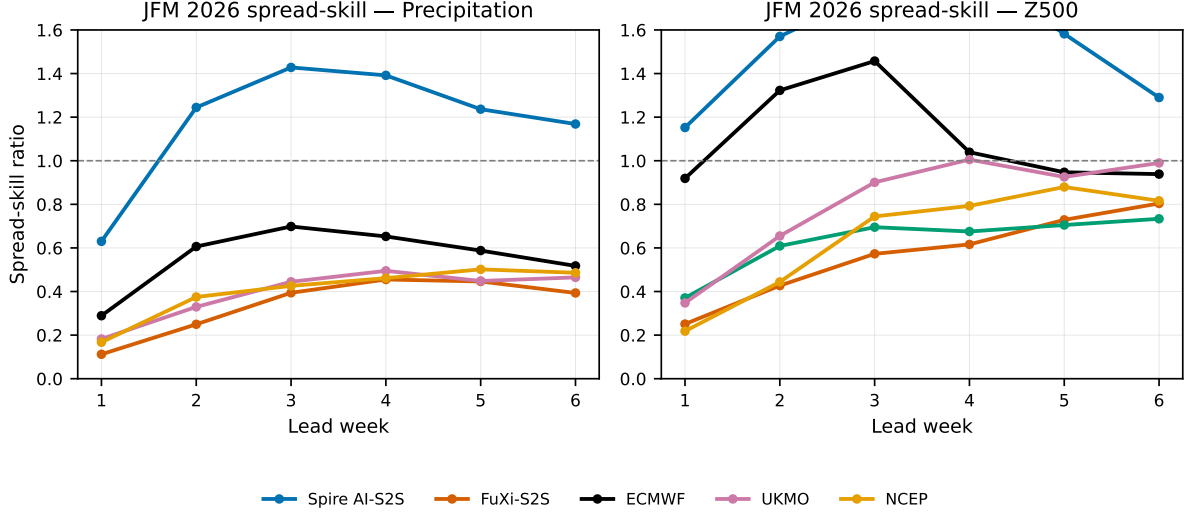


Figure 3: JFM 2026 spread-skill ratio versus lead week; a well-calibrated ensemble lies near unity (dashed line). Values < 1 indicate under-dispersion (overconfidence).

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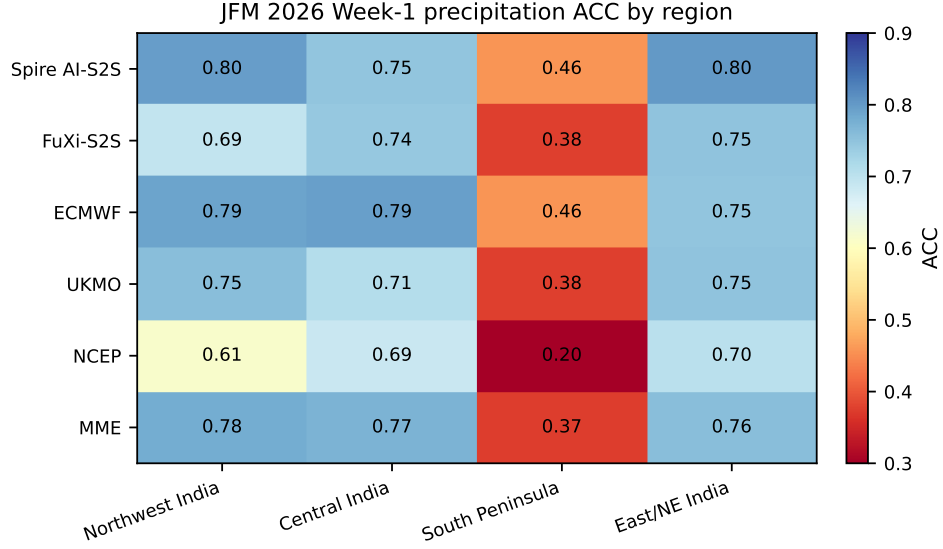


Figure 4: JFM 2026 week-1 precipitation ACC by IMD homogeneous region.

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